ENCODE LONDON 2026 · TECHNICAL DIAGRAM

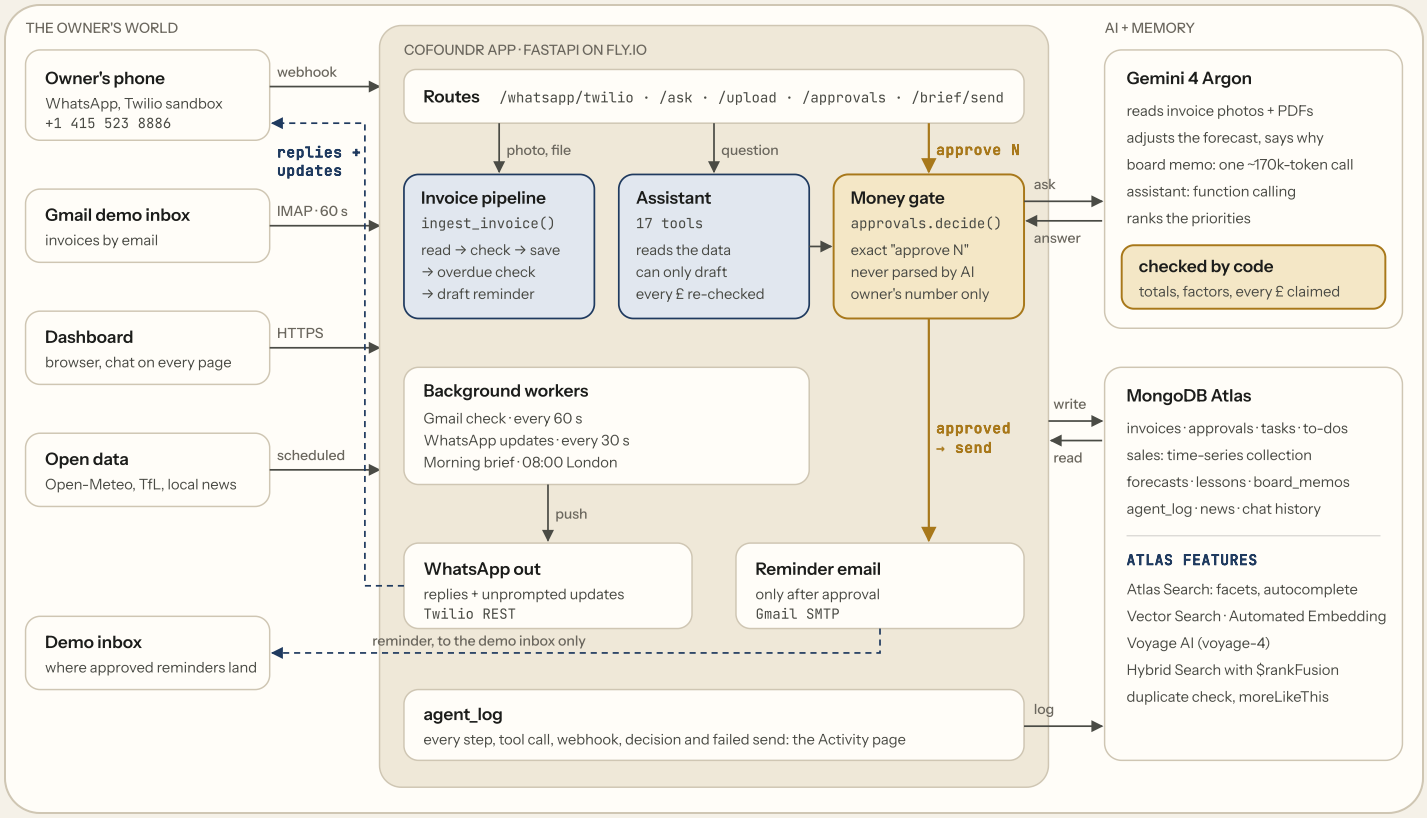
CoFoundr. how it's built

An AI co-founder for a one-truck coffee business. One FastAPI app on Fly.io sits between the owner's channels, Gemini 4 Argon and MongoDB Atlas. It does the work on its own and stops at one gate before any money moves.

The system

Every input enters the same app, every decision passes through `approvals.decide()`, and every step lands in `agent_log`. Gemini is only ever called from inside the app, and its answers are checked by code before they're stored or shown.

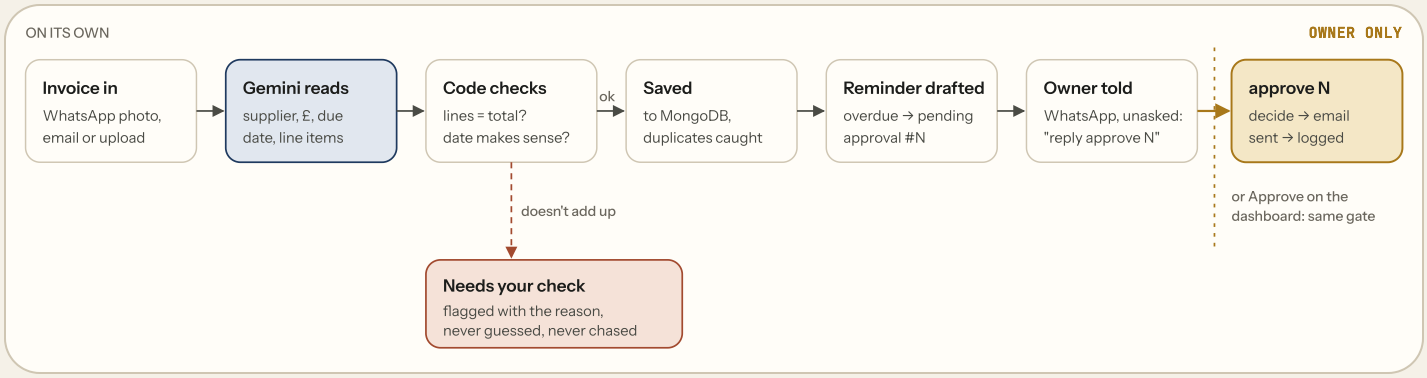
into the app · out to the owner · the money gate



One app owns every connection: the owner's channels never touch the database, and Gemini never talks to the owner directly. The gold path is the only way money moves: an exact "approve N" from the owner's number, or the Approve button, into `approvals.decide()`.

One invoice, start to finish

The path the demo shows. Everything left of the gold line happens on its own; nothing right of it happens without the owner.



Two places the agent stops: a reading that doesn't check out goes to the owner instead of being chased, and nothing past the gold line runs without the owner's exact reply.

Where each part lives

Web app

Routes, dashboard pages, chat panel.

`bridge/main.py · bridge/templates/`

Invoice pipeline

Every input calls this one function.

`backend/pipeline.py · backend/ai/extract.py`

Money gate

The only place a decision is applied.

`backend/approvals.py`

Assistant

Gemini function calling over MongoDB, 17 tools.

`backend/outputs/assistant.py`

WhatsApp

Signed webhook in; replies, updates and the brief out.

`backend/inputs/whatsapp.py · backend/outputs/whatsapp.py`

Search

Atlas Search, Vector Search, Hybrid Search.

`backend/db/search.py`

Forecast + board

Gemini adjusts and grades itself; the 1M-context memo.

`backend/ai/forecast.py · review.py · board.py`

Data contract

Every collection name and document shape.

`backend/db/schema.py`

CoFoundr · Encode London Hackathon 2026 · Sunil & Ceyda. Demo business GAIA Coffee; all business data is synthetic. Live at cofoundr.fly.dev · github.com/sunilramjali/cofoundr